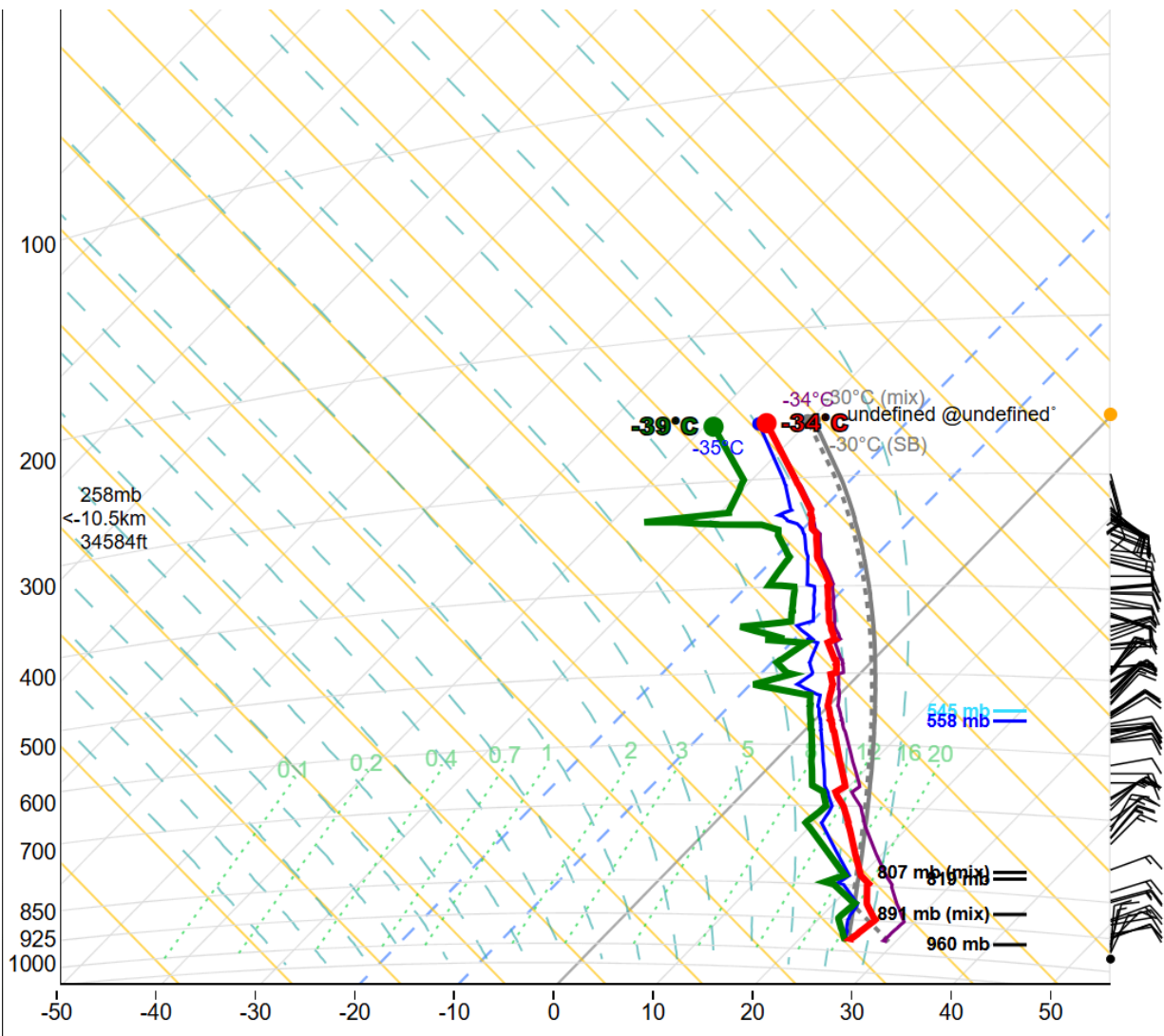


Physics of Atmosphere assignment-Tephigrath and Stability Analysis
Name: Aparna Ramgopal Sharma
Enrollment No: 20243406
Date of Submission: 01/12/24

Tephigrath:

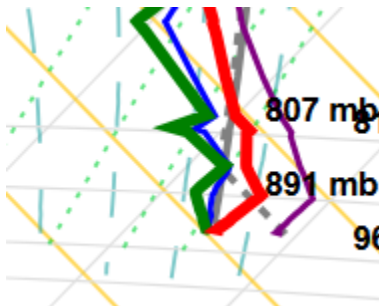
42867 VANP Nagpur Sonegaon Observations at 00Z 23 Sep 2024



Quantities/Level	971 hPa (Surface level)	At 830 hPa (1.4 km)
Mixing Ratio	21.16 g/kg	16.37 g/kg
Saturated Mixing Ratio	22 g/kg	18.5 g/kg
Relative Humidity	96.18%	88.48%
LCL pressure level	970 hPa (30ft)	787 hPa (1.8 km)
LFC pressure level	830 hPa (1.4 km)	711 hPa (2.7 km)
LNB pressure level	711 hPa (2.7 km)	663 hPa (3.3 km)
CCL pressure level	850 hPa (1.2km)	787 hPa (1.8 km)
Convective temperature	36°C	39°C
Dew point depression	1°C	2°C

1. As dew point depression is very low at the surface and it becomes 0°C just 30 ft above the surface this indicated the presence of fog .

2. Dew point depression at 1.4km is 2°C , if the parcel is lifted to LCL which is at 1.8km, condensation occurs, low level clouds can form. As LCL is close to surface these clouds will contain liquid droplets.



3. Till LCL the air parcel follows dry adiabatic lapse rate andn above LCL it follows saturated adiabatic lapse rate. For a surface air parcel, LCL is at just 30 ft above surface, it follows dry adiabatic lapse rate till LCL and then follows saturated adiabatic lapse rate.